MODULE 018 / 100 • STATISTICS & PROBABILITY

# A B Testing and Causal Inference

**Overview:** Sample sizing, power calculations, A/B testing design, and propensity matching.

## 1. Fundamental Architecture & Theoretical Principles

In machine learning engineering and production data science, mastering **A B Testing and Causal Inference** is paramount to designing robust, reproducible, and scalable intelligence systems. This topic anchors foundational understanding across analytical theory, algebraic formulations, and practical implementation.

| Component / Dimension | Technical Specification & Best Practice |
|---|---|
| **Core Formulation** | Governing mechanisms and operational dynamics of A B Testing and Causal Inference. |
| **Production Risk** | Data leakage, numerical instability, excessive latency, or distribution drift. |
| **Standard Remedy** | Robust validation splits, automated testing, regularized optimization, and monitoring. |
| **Primary Frameworks** | PyTorch, Scikit-Learn, Ray, Hugging Face, MLflow, Docker, FastAPI. |

## 2. Engineering Guidelines & Implementation Roadmap

- **Baseline Verification:** Formulate measurable benchmarks and unit tests prior to deployment.
- **Reproducibility:** Pin seed states, containerize runtime environments, and version artifacts.
- **Monitoring:** Instrument low-overhead telemetry to track inference latencies and data quality.
- **Edge Case Containment:** Handle null inputs, out-of-distribution values, and catastrophic drops.

## 3. Competency Checklist & Production Readiness

[ ] Theoretical mastery of mathematical foundations and algorithms for A B Testing and Causal Inference.

[ ] Pipeline implementation with leak-free preprocessing and cross-validation.

[ ] Benchmarking evaluation completed against industry standard baselines.

[ ] Production deployment artifacts packaged with automated tests.